Factors, Multiples and Divisibility

Australian Curriculum v9.0 Year 5 Maths
AC9M5N02

Name: ______________________________ Date: __________________

LEARNING TARGET

I can express numbers as products of factors, recognise multiples and test divisibility.

KEY STRATEGY + WORKED EXAMPLE

Use factor pairs systematically from 1 upward. A number is divisible by another when the quotient is a whole number with no remainder.

Example: 36 = 1 x 36 = 2 x 18 = 3 x 12 = 4 x 9 = 6 x 6. Therefore its factors are 1,2,3,4,6,9,12,18,36.

A. TRY TOGETHER - USE THE STRATEGY

1. List the factor pairs of 24.
2. Write the first five multiples of 7.
3. Is 135 divisible by 5? Explain.
4. Is 246 divisible by 3? Use a divisibility test.

B. CORE PRACTICE - SHOW YOUR THINKING

5. List all factors of 18.

6. List all factors of 40.

7. Write 72 as three different products.

8. First six multiples of 9.

9. Common factors of 24 and 36.

10. Lowest common multiple of 4 and 6.

11. Is 418 divisible by 2, 3, 5 or 10?

12. Find a factor of 84 between 8 and 15.

C. INDEPENDENT PRACTICE - CALCULATE AND CHECK

13. Factors of 49

14. Multiples of 8 below 50

15. Is 561 divisible by 3?

16. Is 2,340 divisible by 10?

17. Greatest common factor of 18 and 30

18. Lowest common multiple of 5 and 8

19. Complete: 96 = 12 x ___

20. Complete: ___ x 15 = 120

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D. ANALYSE & APPLY

21. A teacher arranges 48 students in equal rows. List every possible row size greater than 1 and less than 48.

22. Two bells ring every 6 and 8 minutes. If they ring together now, when will they next ring together?

23. Correct Ava: 'Every multiple of 6 is also a multiple of 12.' Give evidence.

E. REASON, CHECK AND CORRECT

24. Explain the difference between a factor and a multiple.

25. A number is divisible by 2 and 5 and lies between 90 and 130. List all possibilities.

EXIT TICKET + CHALLENGE

26. Find all common factors of 36 and 54 and identify the greatest common factor.

Challenge: Create a number with exactly six factors. Prove that your list is complete.

F. STUDENT REVIEW - SHOW WHAT YOU UNDERSTAND

Use your own words so your teacher can see what you understand and what to practise next.
27. Explain one divisibility test you used.

28. Check one factor list by multiplying every pair.

29. Circle one: I understand it | I need more practice | I need help

30. The easiest part was: ____

31. The most difficult part was: ____

Why was it difficult? ____

32. My next practice step is: ____

PARENT / TEACHER TIP

Ask your child to prove one factor using division and one multiple using multiplication. Encourage a complete, ordered factor list.

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TEACHER COPY - ANSWERS AND SUPPORT

ANSWER KEY

A. 1) 1x24, 2x12, 3x8, 4x6. 2) 7,14,21,28,35. 3) Yes; it ends in 5. 4) Yes; digit sum is 12.

B. 5) 1,2,3,6,9,18. 6) 1,2,4,5,8,10,20,40. 7) e.g. 1x72, 2x36, 3x24. 8) 9,18,27,36,45,54. 9) 1,2,3,4,6,12. 10) 12. 11) 2 only. 12) 12.

C. 13) 1,7,49. 14) 8,16,24,32,40,48. 15) Yes. 16) Yes. 17) 6. 18) 40. 19) 8. 20) 8.

D. 21) 2,3,4,6,8,12,16,24. 22) 24 minutes. 23) False; 6 is a multiple of 6 but not 12.

E. 24) A factor divides a number exactly; a multiple results from multiplying it. 25) 100,110,120.

EXIT. 26) 1,2,3,6,9,18; GCF = 18.

Challenge: Answers vary. Check that the student's example follows every condition and that the reasoning is mathematically correct.

F. 27-28) Answers vary. Look for a clear strategy and a valid second check. 29-32) Use the reflection to identify confidence, difficulty and the next teaching step.

COMMON MISCONCEPTIONS

Students may confuse factors with multiples or stop before finding every factor pair. Record pairs in order until they meet.

TEACHER CHECK

[ ] Uses an appropriate Year 5 Maths strategy.

[ ] Shows or explains thinking clearly.

[ ] Checks a response using evidence, a rule, or a second method.

[ ] Identifies a specific difficulty and useful next practice step.

PARENT / TEACHER TIP

Ask the child to explain how they know, not only state an answer. If the explanation is unclear, use small objects, a drawing, or a number line before returning to symbols.

NEXT STEP

Revisit the item the child marked difficult. Model one example, solve one together, then ask the child to complete one independently and explain the strategy.